EX 7: ANALOG BEAMFORMING ANALYSIS USING MATLAB

Aim:

To analyze the performance of analog beamforming by studying beamforming gain, beam width, and main lobe direction using MATLAB.

Objective 1:MATLAB Code and output

clc;

clear;

close all;

antennas = [2 4 8 16 32]; % Number of Antenna Elements

gain = 10*log10(antennas); % Beamforming Gain (dB)

figure

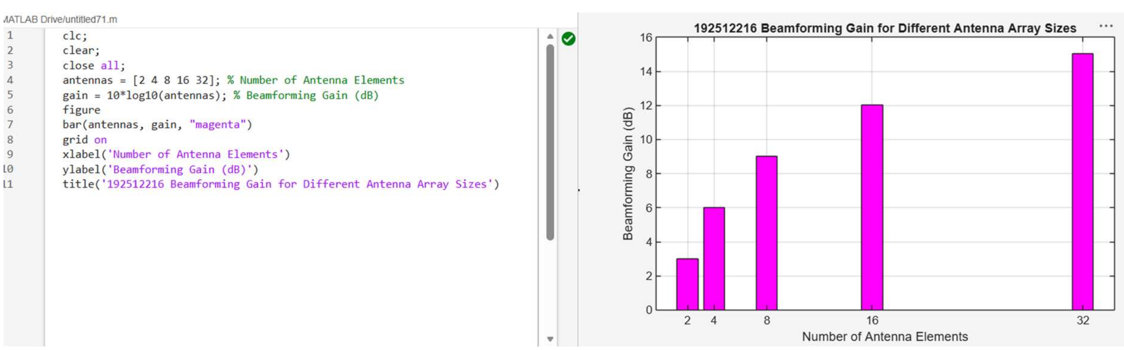
bar(antennas, gain)

grid on

xlabel('Number of Antenna Elements')

ylabel('Beamforming Gain (dB)')

title('Beamforming Gain for Different Antenna Array Sizes')



Objective 2: MATLAB Code and Output

clc;

clear;

close all;

antennas = [2 4 8 16 32]; % Number of Antenna Elements

beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)

figure

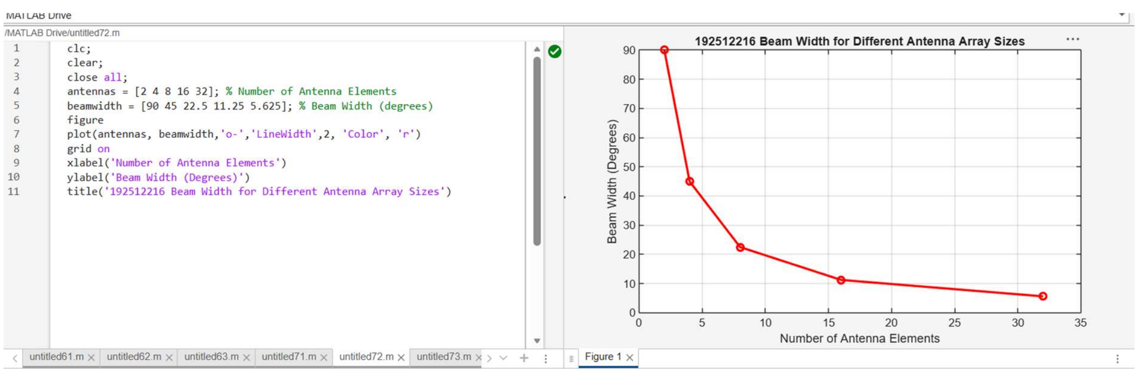
plot(antennas, beamwidth,'o-','LineWidth',2)

grid on

xlabel('Number of Antenna Elements')

ylabel('Beam Width (Degrees)')

title('Beam Width for Different Antenna Array Sizes')



Objective 2: MATLAB Code and Output

clc;

clear;

close all;

theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)

gain = [1 1 1 1 1]; % Constant normalized gain

figure

polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)

title('Main Lobe Direction in Analog Beamforming')

Result:

The MATLAB analysis successfully demonstrated the characteristics of analog beamforming. The results showed that increasing the number of antenna elements improves beamforming gain and reduces beam width, while phase adjustment enables the main lobe to be steered toward different directions.